

Intrinsic Bulk-Square Transfer and the Shrinking-Disk Determinant Gate

Diagonal Small-Noise Limits, Trace-Ideal Continuity, and the Fixed-Disk Exponential
Barrier

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Abstract

RH-78 showed conditionally that either of two all-level Hardy corridors would close intrinsic identification on every strict mesh schedule $n\sigma^2 \rightarrow \infty$. This paper determines exactly how far that conclusion propagates toward the two-step Fredholm determinant.

Let $B_{n,\sigma}^{\text{int}}$ be the intrinsic finite bulk and B_σ the continuum-anchored bulk. If

$$\|B_{n,\sigma}^{\text{int}} - B_\sigma\|_2 \leq \varepsilon, \quad \|B_\sigma\|_2 \leq M,$$

then

$$\|(B_{n,\sigma}^{\text{int}})^2 - B_\sigma^2\|_1 \leq \delta := \varepsilon(2M + \varepsilon).$$

For the Fredholm determinants on $|w| \leq R$,

$$|D_{n,\sigma}(w) - D_\sigma(w)| \leq R\delta \exp\{1 + RM^2 + R(M + \varepsilon)^2\}.$$

Under RH-78's conditional polylogarithmic identification and the Gaussian bulk size $M = O(\sigma^{-1/2})$, the square error tends to zero along every strict $n\sigma^2 \rightarrow \infty$ diagonal. The determinant bound is uniform on shrinking disks $R = O(\sigma)$, where the exponential remains $O(1)$.

On a fixed disk, however, the same standard estimate contains $\exp(O(R/\sigma))$. It therefore does not transfer the small-noise limit, even though the trace-norm square error vanishes. This is a proof-method barrier, not a theorem of determinant divergence.

A 256-bit Arb audit uses the RH-78 stress diagonal and the conservative bulk bound $M \leq 1.55\sigma^{-1/2}$. The trace-norm square upper decreases from 0.5754 to 0.09180. On $R = 0.01\sigma$, the determinant upper decreases from 2.63×10^{-3} to 2.62×10^{-5} . On the fixed disk $R = 0.01$, the standard bound bottoms near 1.57×10^{-2} and then worsens to 0.3051.

Thus conditional Stage A4 reaches trace-norm squares and shrinking-disk determinants. Entry to Stage A5 on fixed disks requires pole renormalization or a sharper relative determinant argument; additional finite precision is irrelevant to this exponential obstruction. A5, Hilbert–Polya, and the Riemann Hypothesis remain open.

Keywords: trace ideal; Fredholm determinant; diagonal limit; intrinsic bulk; pole renormalization; small noise.

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1 Introduction

At fixed positive noise, RH-45 proved Hilbert–Schmidt bulk convergence, trace-norm convergence of squares, and local uniform convergence of the two-step determinant [5]. RH-46 then showed that the small-noise determinant germ has a genuine double-pole obstruction and that unrenormalized fixed-disk convergence is not the correct endpoint [4].

RH-78 now supplies a conditional diagonal identification bound [3]. It is natural to ask whether this alone permits the finite intrinsic determinant to be exchanged with the continuum and small-noise limits. The answer splits cleanly:

1. bulk squares transfer in trace norm;
2. determinants transfer on disks shrinking like σ ;
3. the generic fixed-disk determinant estimate is exponentially nonuniform and cannot open A5.

2 Hilbert–Schmidt identification to trace-class squares

The following standard estimate is the bridge from Stage A4 to two-step trace ideals.

Theorem 2.1 (Intrinsic square transfer). *Let $B, \tilde{B} \in \mathfrak{S}_2$ satisfy $\|\tilde{B} - B\|_2 \leq \varepsilon$ and $\|B\|_2 \leq M$. Then*

$$\|\tilde{B}\|_2 \leq M + \varepsilon, \quad (1)$$

$$\|\tilde{B}^2 - B^2\|_1 \leq \varepsilon(2M + \varepsilon) =: \delta. \quad (2)$$

Proof. Use $\tilde{B}^2 - B^2 = (\tilde{B} - B)\tilde{B} + B(\tilde{B} - B)$ and $\|XY\|_1 \leq \|X\|_2 \|Y\|_2$. $\square$

Under RH-78's conditional corridor theorem,

$$\varepsilon_{n,\sigma} = O\left(n^{-2}\sigma^{-13/4}L(\sigma)\right), \quad (3)$$

where L is polylogarithmic. The Gaussian bulk has $M_\sigma = O(\sigma^{-1/2})$. Hence

$$\delta_{n,\sigma} = O\left(n^{-2}\sigma^{-15/4}L(\sigma)\right) + O(\varepsilon_{n,\sigma}^2). \quad (4)$$

Proposition 2.2 (Strict diagonal trace-norm closure). *If $n\sigma^2 \rightarrow \infty$, then $\delta_{n,\sigma} \rightarrow 0$.*

Proof. Write $n = \sigma^{-2}G(\sigma)$ with $G \rightarrow \infty$. The leading term in (4) is $O(\sigma^{1/4}G^{-2}L)$; every fixed polylogarithm is dominated by $\sigma^{1/4}$. $\square$

Thus conditional intrinsic identification is strong enough for even traces and bulk-square trace-class convergence.

3 Determinant disks

Define

$$D_{n,\sigma}(w) = \det(I - w(B_{n,\sigma}^{\text{int}})^2), \quad \text{quad}D_\sigma(w) = \det(I - wB_\sigma^2).$$

The standard trace-ideal determinant continuity estimate [2] gives:

Theorem 3.1 (Determinant disk transfer). *Under Theorem 2.1, for every $R \geq 0$,*

$$\sup_{|w| \leq R} |D_{n,\sigma}(w) - D_\sigma(w)| \leq R\delta \exp(1 + RM^2 + R(M + \varepsilon)^2). \quad (5)$$

Proof. Apply $|\det(I + C) - \det(I + D)| \leq \|C - D\|_1 e^{1 + \|C\|_1 + \|D\|_1}$ with $C = -w\tilde{B}^2$, $D = -wB^2$, and use $\|B^2\|_1 \leq \|B\|_2^2$. $\square$

Corollary 3.2 (Shrinking-disk diagonal transfer). *If $R_\sigma \leq \rho\sigma$ and $n\sigma^2 \rightarrow \infty$, then the right side of (5) tends to zero.*

Proof. Since $M^2 = O(\sigma^{-1})$, the exponential is $O_\rho(1)$. The remaining factor is $O(\sigma\delta)$, which tends to zero by Proposition 2.2. $\square$

Remark 3.3 (Fixed-disk barrier). *For fixed $R > 0$, (5) contains $\exp(O(R/\sigma))$. No power or polylogarithmic decay in $\delta_{n,\sigma}$ controls that exponential on the minimal strict mesh schedule. This does not prove that $D_{n,\sigma} - D_\sigma$ diverges; it proves that generic absolute determinant continuity is the wrong fixed-disk tool.*

4 Five-anchor Arb audit

The audit uses RH-78's stress diagonal $n = \sigma^{-2}(k+2)$, its conditional identification ball, and

$$M_\sigma \leq 1.55\sigma^{-1/2}. \quad (6)$$

All scalar products, exponentials, and disk bounds are evaluated at 256-bit Arb precision [1].

Table 1: Conditional square and determinant transfer uppers.

σ	ε	square error	$R = 0.01\sigma$	$R = 0.01$
0.16	0.07355	0.5754	$2.63 \cdot 10^{-3}$	0.02124
0.08	0.02867	0.3150	$7.19 \cdot 10^{-4}$	0.01566
0.04	0.01268	0.1966	$2.25 \cdot 10^{-4}$	0.01780
0.02	0.005997	0.1315	$7.51 \cdot 10^{-5}$	0.03955
0.01	0.002961	0.09180	$2.62 \cdot 10^{-5}$	0.3051

The shrinking-disk upper decreases monotonically by two orders of magnitude. The fixed-disk upper first improves and then deteriorates rapidly as the trace-norm exponential takes over.

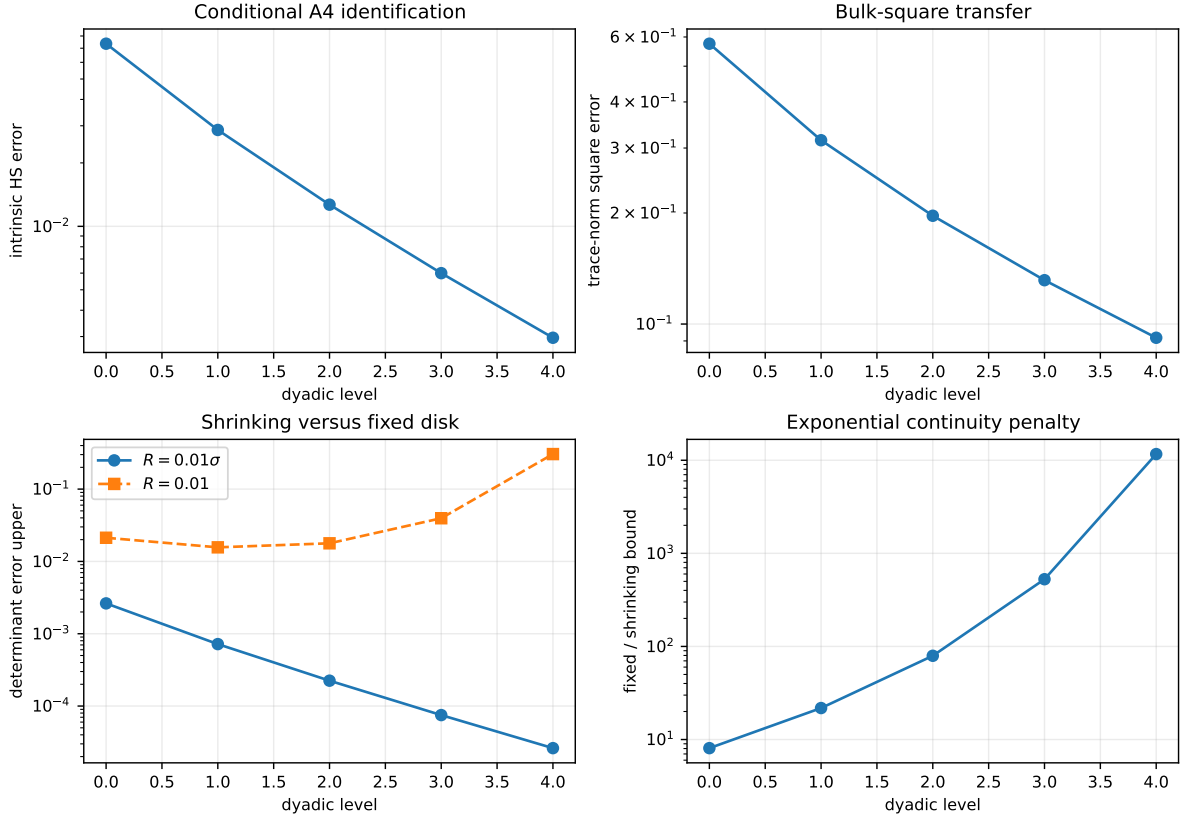


Figure 1: Conditional intrinsic error, square trace-norm transfer, shrinking versus fixed determinant disks, and the exponential penalty ratio.

5 Route consequence

The Stage-A output is now sufficient for trace-class bulk squares along the small-noise diagonal. It also matches RH-46's natural shrinking coordinate $w = O(\sigma)$. What it does not supply is a nontrivial fixed-disk limit.

Stage A5 must therefore alter the object before taking the limit. The two remaining candidates are:

1. divide out the explicit deterministic double-pole factor and prove a pole-renormalized determinant family is locally bounded;
2. construct a relative determinant whose continuity depends on a renormalized trace norm rather than $\|B_\sigma^2\|_1 = O(\sigma^{-1})$.

RH-80 tests this entry gate.

This paper is conditional on an all-level RH-78 corridor. It does not close Stage A1, unconditional Stage A4, or Stage A5; construct a Hilbert–Polya operator; prove a $T \log T$ law or prime-power trace formula; identify zeta zeros; or prove the Riemann Hypothesis.

6 Conclusion

Intrinsic identification survives squaring and the strict diagonal limit. Determinants also transfer, but only on the shrinking scale naturally selected by the double-pole geometry. The fixed-disk exponential barrier makes the next move unambiguous: renormalize first, then seek a limit.

References

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